STRUCTURE → NAME

A01 | Foundation · Molecular formula: C_6H_{14}

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 5 carbon atoms → pentane.

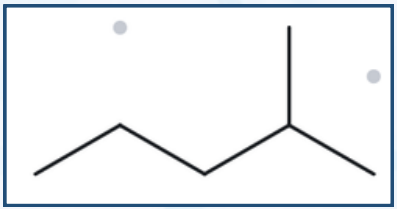
Numbering: Left → right gives the methyl substituent at 4; Right → left gives the methyl substituent at 2; First point of difference: 2 < 4, so number from right → left.

Substituent: One methyl group (–CH_3) attached to 2 → 2-methyl.

3) My Name

2-methylpentane

4) The Alkane Check



2-methylpentane

✓ **Match**

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A02 | Foundation · Molecular formula: C_7H_{16}

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 6 carbon atoms → hexane.

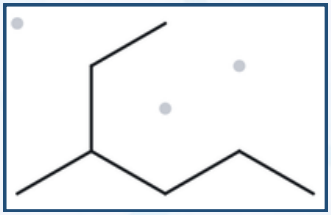
Numbering: Upper end → right end gives 3; Right end → upper end gives 4; First point of difference: 3 < 4, so number from the upper end.

Substituent: One methyl group attached to C3 → 3-methyl.

3) My Name

3-methylhexane

4) The Alkane Check



3-methylhexane

✓ **Match**

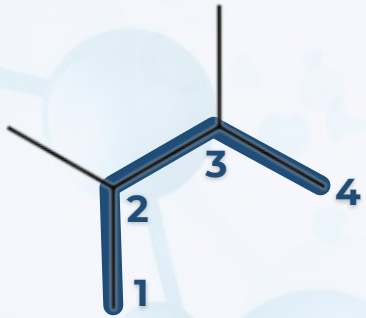
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A03 | Foundation · Molecular formula: C_6H_{14}

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 4 carbon atoms → butane.

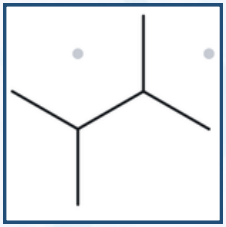
Numbering: Left → right gives methyl groups at 2 and 3; Right → left also gives methyl groups at 2 and 3; Both directions give the same locants.

Substituent: Two methyl groups attached to 2 and 3 → 2,3-dimethyl.

3) My Name

2,3-dimethylbutane

4) The Alkane Check



2,3-dimethylbutane

✓ **Match**

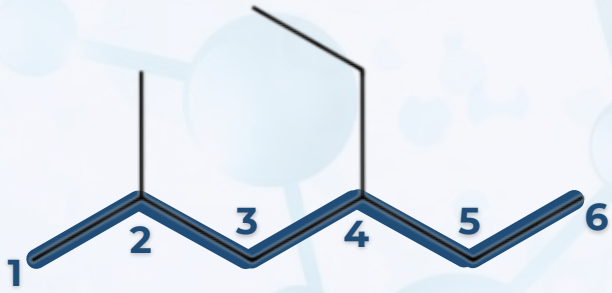
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A04 | Developing · Molecular formula: C_9H_{20}

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 6 carbon atoms → hexane. There are multiple 6-carbon paths, but they give the same substituents when numbered correctly.

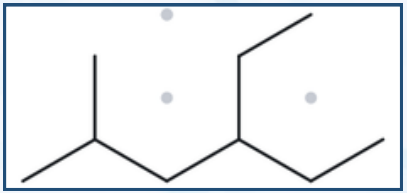
Numbering: Left → right gives substituents at 2 and 4; Right → left gives substituents at 3 and 5; First point of difference: 2 < 3, so number from the left.

Substituent: Methyl group at 2; Ethyl group at 4; Arrange substituents alphabetically: ethyl before methyl.

3) My Name

4-ethyl-2-methylhexane

4) The Alkane Check



4-ethyl-2-methylhexane

✓ **Match**

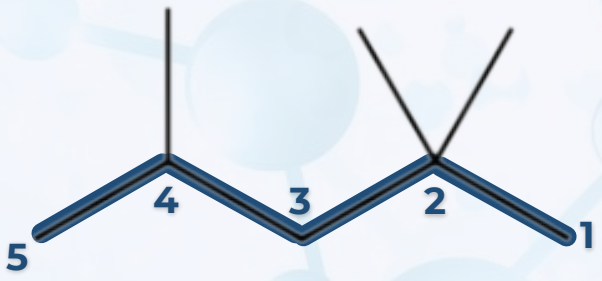
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A05 | Developing · Molecular formula: C_8H_{18}

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 5 carbon atoms → pentane.

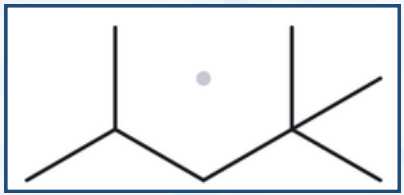
Numbering: Left → right gives methyl groups at 2, 4, and 4; Right → left gives methyl groups at 2, 2, and 4; First point of difference: 2,2,4 is lower than 2,4,4, so number from the right.

Substituent: Three methyl groups attached to 2, 2, and 4 → 2,2,4-trimethyl.

3) My Name

2,2,4-trimethylpentane

4) The Alkane Check



2,2,4-trimethylpentane

✓ **Match**

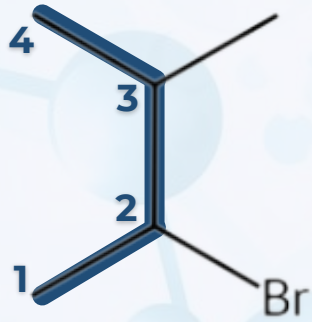
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A06 | Developing · Molecular formula: $C_5H_{11}Br$

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 4 carbon atoms → butane.

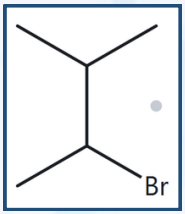
Numbering: Starting from the upper-left end gives methyl at 2 and bromo at 3; Starting from the lower-left end gives bromo at 2 and methyl at 3; Both directions give the locant set 2,3; First point of difference (alphabetical tie-break): bromo comes before methyl, so bromo receives the lower number.

Substituent: Bromo (–Br) on 2; Methyl ($–CH_3$) on 3.

3) My Name

2-bromo-3-methylbutane

4) The Alkane Check



2-bromo-3-methylbutane

✓ **Match**

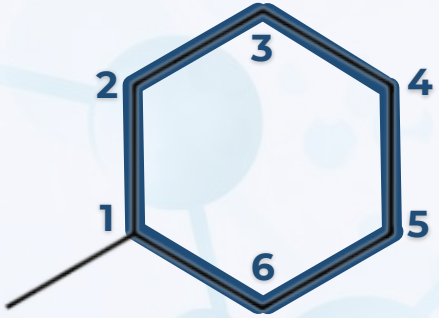
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A07 | Developing · Molecular formula: C_7H_{14}

1) Numbered Parent



2) Naming Trail

Parent: Six-membered carbon ring = cyclohexane.

Numbering: Start at the ring carbon attached to the methyl group (1); Continue clockwise or counterclockwise; Since there is only one substituent, either direction gives the same numbering.

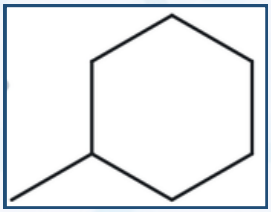
Substituent: One methyl group ($–CH_3$) attached to 1 → 1-methyl.

3) My Name

methylcyclohexane

Note: The locant 1 is unnecessary because there is only one substituent on the ring.

4) The Alkane Check



methylcyclohexane

✓ **Match**

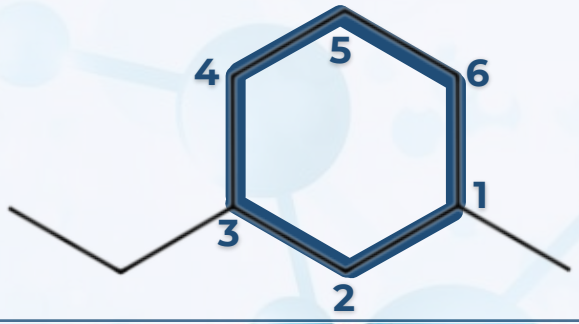
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A08 | Challenge · Molecular formula: C_9H_{18}

1) Numbered Parent



2) Naming Trail

Parent: Longest parent structure = six-carbon ring → cyclohexane. The ring is the parent because it contains more carbon atoms than either attached side chain.

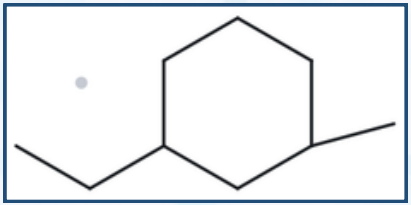
Numbering: Start at the ring carbon attached to the ethyl group; Number toward the methyl group to obtain the lowest locants; Clockwise: ethyl at 1 and methyl at 3; Counterclockwise: ethyl at 1 and methyl at 5; First point of difference: 3 < 5, so choose the first direction.

Substituent: Ethyl group (–CH_2CH_3) on 1; Methyl group (–CH_3) on 3; Arrange alphabetically: ethyl before methyl.

3) My Name

1-ethyl-3-methylcyclohexane

4) The Alkane Check



1-ethyl-3-methylcyclohexane

✓ **Match**

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A09 | Challenge · Molecular formula: $C_{10}H_{22}$

1) Numbered Parent



2) Naming Trail

Parent: Longest continuous chain = 7 carbon atoms → heptane.

Numbering: Left → right gives the substituent at 4; Right → left also gives the substituent at 4; Both directions give the same locant, so either direction is acceptable.

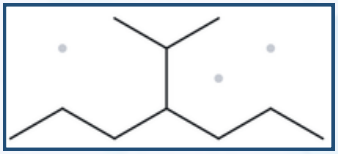
Substituent: One propan-2-yl (isopropyl) group attached to 4. The branch contains: 1 central carbon attached to the parent chain and 2 methyl groups attached to that central carbon.

3) My Name

4-(propan-2-yl)heptane

Commonly written as:
4-isopropylheptane

4) The Alkane Check



2-methyl-3-propylhexane

✗ Mismatch

The correct name is **4-(propan-2-yl)heptane**. The Alkane's mistake was selecting a six-carbon parent chain instead of the longest available seven-carbon chain. According to the IUPAC rule, the longest continuous carbon chain must be selected as the parent before identifying substituents. Therefore, 4-(propan-2-yl)heptane is correct.

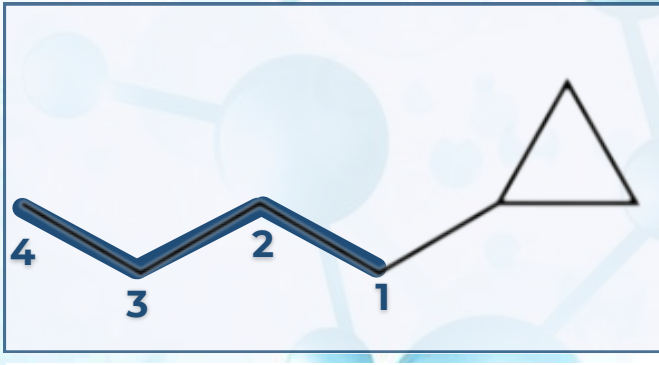
VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

STRUCTURE → NAME

A10 | Challenge · Molecular formula: C_7H_{14}

1) Numbered Parent



2) Naming Trail

Parent: Longest acyclic chain = 4 carbon atoms → butane. The chain is chosen as the parent because it contains more carbon atoms than the cyclopropane ring (4 vs. 3).

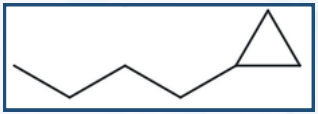
Numbering: Left → right gives the cyclopropyl substituent at 4; Right → left gives the cyclopropyl substituent at 1; Choose right → left because 1 is lower than 4.

Substituent: One cyclopropyl group attached to 1 → 1-cyclopropyl.

3) My Name

1-cyclopropylbutane

4) The Alkane Check



butylcyclopropane

✗ Mismatch

Both 1-cyclopropylbutane and butylcyclopropane refer to the same structure, but they differ in their choice of parent structure and naming. The correct IUPAC name is **1-cyclopropylbutane** because the longest acyclic chain contains four carbon atoms, while the cyclopropane ring contains only three. According to the parent structure selection rule, the acyclic chain is chosen as the parent when it contains more carbon atoms than the ring; thus, butane is the parent and cyclopropyl is the substituent

VERIFICATION STRIP

- ☑ longest chain confirmed
- ☑ lowest locants
- ☑ alphabetical order
- ☑ punctuation
- ☑ formula counted

NAME → STRUCTURE

B01 | Foundation · Given Name: 2-methylpropane

1) Name Decoded

Parent: Propane → a 3-carbon chain.

Substituent: Methyl on 2.

Expected Formula: 3 + 1 = 4 C; acyclic alkane → C_4H_{10}.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing: C_4H_{10}

Acyclic alkane: C_nH_{2n+2} = $C_4H_{2(4)+2}$ = C_4H_{10} ✓

4) Back-Check

2-methylpropane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B02 | Foundation · Given Name: 3-ethylpentane

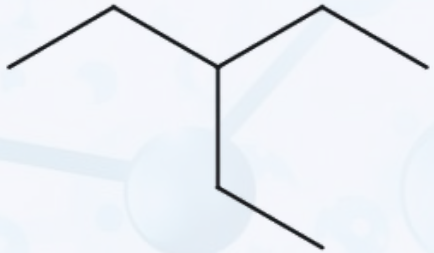
1) Name Decoded

Parent: Pentane → a 5-carbon chain.

Substituent: Ethyl on 3.

Expected Formula: 5 + 2 = 7 C; acyclic alkane → C_7H_{16}.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing: C_7H_{16}

Acyclic alkane: C_nH_{2n+2} = $C_7H_{2(7)+2}$ = C_4H_{16} ✓

4) Back-Check

3-ethylpentane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B03 | Foundation · Given Name: 2,2-dimethylbutane

1) Name Decoded

Parent: Butane → a 4-carbon chain.

Substituent: Two methyl groups on 2

Expected Formula: 4 + 2 = 6 C; acyclic alkane → C_6H_{14}.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing: C_6H_{14}

Acyclic alkane: C_nH_{2n+2} = $C_6H_{2(6)+2}$ = C_6H_{14} ✓

4) Back-Check

2,2-dimethylbutane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B04 | Developing · Given Name: 3-ethyl-2-methylpentane

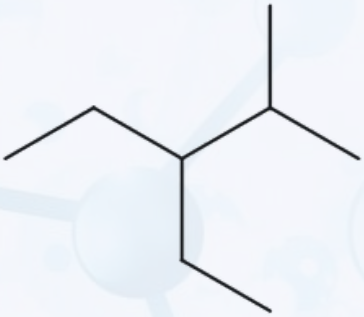
1) Name Decoded

Parent: Pentane → a 5-carbon chain.

Substituent: Ethyl on 3; methyl on 2.

Expected Formula: 5 + 2 + 1 = 8 C; acyclic alkane → C_8H_{18}.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing: C_8H_{18}

Acyclic alkane: C_nH_{2n+2} = $C_8H_{2(8)+2}$ = C_8H_{18} ✓

4) Back-Check

3-ethyl-2-methylpentane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B05 | Developing · Given Name: 1-chloro-2-methylpropane

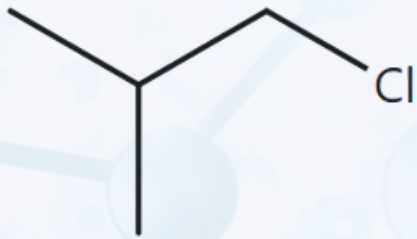
1) Name Decoded

Parent: Propane → a 3-carbon chain.

Substituent: Chloro on 1; methyl on 2.

Expected Formula: 3 + 1 = 4 C; acyclic alkane → C_4H_{10}; one H replaced by Cl → C_4H_9Cl.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing: C_4H_9Cl

Acyclic alkane: C_nH_{2n+2} = $C_4H_{2(4)+2}$ = C_4H_{10}

One hydrogen is replaced by chlorine: C_4H_9Cl ✓

4) Back-Check

1-chloro-2-methylpropane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B06 | Developing · Given Name: 1,2-dimethylcyclopentane

1) Name Decoded

Parent: Cyclopentane → a 5-carbon ring

Substituent: Methyl on 1; methyl on 2

Expected Formula: 5 + 1 + 1 = 7 C; one ring → C_7H_{14}; two methyl substitutions replace two H atoms with two CH_3 groups, giving C_7H_{14}.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing:
C_7H_{14}

Cycloalkane with one ring:
$C_nH_{2n} = C_7H_{2(7)} = C_7H_{14}$ ✓

4) Back-Check

1,2-dimethylcyclopentane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B07 | Developing · Given Name: 4-ethyl-3,3-dimethylheptane

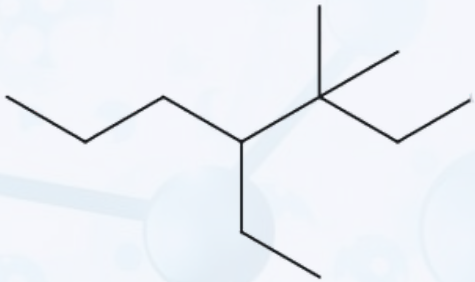
1) Name Decoded

Parent: Heptane → a 7-carbon chain.

Substituent: Ethyl on 4; two methyl groups on 3.

Expected Formula: 7 + 2 + 1 + 1 = 11 C; acyclic alkane → $C_{11}H_{24}$.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing: $C_{11}H_{24}$

Acyclic alkane: C_nH_{2n+2} = $C_{11}H_{2(11)+2}$ = $C_{11}H_{24}$ ✓

4) Back-Check

4-ethyl-3,3-dimethylheptane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B08 | Challenge · Given Name: tert-butylcyclohexane

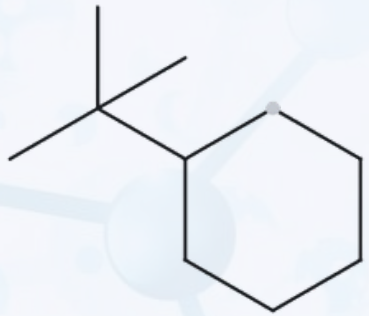
1) Name Decoded

Parent: Cyclohexane → a 6-carbon ring.

Substituent: tert-Butyl on C1 → a branched substituent containing 4 carbon atoms (one central carbon bonded to three methyl groups).

Expected Formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing:
$C_{10}H_{20}$

Cycloalkane with one ring:
$C_nH_{2n} = C_{10}H_{2(10)} = C_{10}H_{20}$ ✓

4) Back-Check

(1,1-dimethylethyl)cyclohexane

✓ **Same as the given name**

Note: The tert-butylcyclohexane and (1,1-dimethylethyl)cyclohexane are the exact same compound because they describe the identical six-carbon ring with a four-carbon branch using two different official naming systems. While "tert-butyl" is the widely accepted common shorthand nickname for this specific four-carbon branch, "(1,1-dimethylethyl)" is its systematic IUPAC blueprint name that breaks down the branch step-by-step as a two-carbon chain (ethyl) with two one-carbon branches (1,1-dimethyl) attached to it.

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B09 | Challenge · Given Name: 3-ethyl-1,1-dimethylcyclohexane

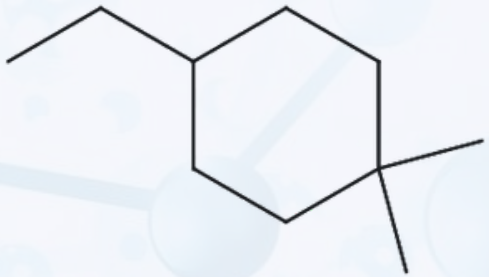
1) Name Decoded

Parent: Cyclohexane → a 6-carbon ring.

Substituent: Ethyl on 3; two methyl groups on 1.

Expected Formula: 6 + 2 + 1 + 1 = 10 C; one ring → $C_{10}H_{20}$.

2) Bond-Line Structure



3) Formula Check

Counted from the drawing:
$C_{10}H_{20}$

Cycloalkane with one ring:
$C_nH_{2n} = C_{10}H_{2(10)} = C_{10}H_{20}$ ✓

4) Back-Check

4-ethyl-1,1-dimethylcyclohexane

✓ **Same as the given name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded

NAME → STRUCTURE

B10 | Challenge · Given Name: 2-ethylbutane *(incorrect name)*

The correct IUPAC name is **3-methylpentane**, not 2-ethylbutane, because the longest continuous carbon chain contains five carbon atoms, making pentane the parent chain.

1) Name Decoded

Parent:

Substituent:

Expected Formula:

2) Bond-Line Structure



3) Formula Check

Counted from the drawing:
C_8H_{18}

Acyclic alkane: C_nH_{2n+2} =
$C_8H_{2(8)+2} = C_8H_{18}$ ✓

4) Back-Check

3-methylpentane

✓ **Same as the resolved name**

VERIFICATION STRIP

- ☑ parent length matches
- ☑ substituent positions matched
- ☑ formula agrees
- ☑ back-check recorded